

ECON100B - SECTION 7

The Solow Growth Model Pt. 2, PSET Questions

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 1 due (Tomorrow!) Friday, 2/13
- PSET 2 due Friday, 2/27

Agenda

- Solow Growth Model Wrap Up
- PSET review and questions

Question of the Day

1. We've spent the past few classes discussing the Cobb-Douglas and Solow models. While these models do a good job of explaining economic output, they are also imperfect models. What would you consider some of the flaws of the Cobb-Douglas and Solow models?
2. What is your favorite season/month of the year?

QOTD Solutions

1. No correct answer, up to your perspective, but here are a few thoughts:

- The Cobb-Douglas and Solow models are logical and do a good job of explaining output
- However, there is a heavy reliance on “A”, or Total Factor Productivity, which is not something that we can directly measure
- The Cobb-Douglas and Solow models assume consistently diminishing returns to capital, which might not play out in empirical evidence (too simple of a model?) – convergence not observed empirically
- All capital and all labor is assumed to be substitutable for each other – is this really the case?

At the end of the day, our models don't need to be (and never will be) a perfect map to reality. Both the Cobb Douglas and Solow models help us understand and build intuition about the economic factors that drive output and economic growth

The Solow Growth Model (Wrap up)



Solow Model Setup Review

1. What is the formula for per-capita output under the Solow Model. How does it show up on the graph?
2. What is the formula for change in capital per person under the Solow Model? How do these two components show up on the graph?
3. What does the Solow Model say about accumulation of capital in the long run?
4. How are growth rates different for a country that is far away from reaching its steady state versus close to reaching its steady state? This concept is known as Transition Dynamics

The Steady state

- While we can't solve for capital at any given point, we are able to solve for the steady state level of capital:

$$\bar{s}y^* = \bar{d}k^*.$$

$$\bar{s}\bar{A}k^{*1/3} = \bar{d}k^*.$$

$$k^* = \left(\frac{\bar{s}\bar{A}}{\bar{d}} \right)^{3/2}.$$

Steady State Output

- Associated with a steady-state level of capital is a steady state level of production

$$y^* = \bar{A}k^{*1/3}.$$

- Thus, we achieve the following expression for steady-state production if we plug in k^* into the equation:

$$y^* = \left(\frac{\bar{s}}{\bar{d}}\right)^{1/2} \bar{A}^{3/2}.$$

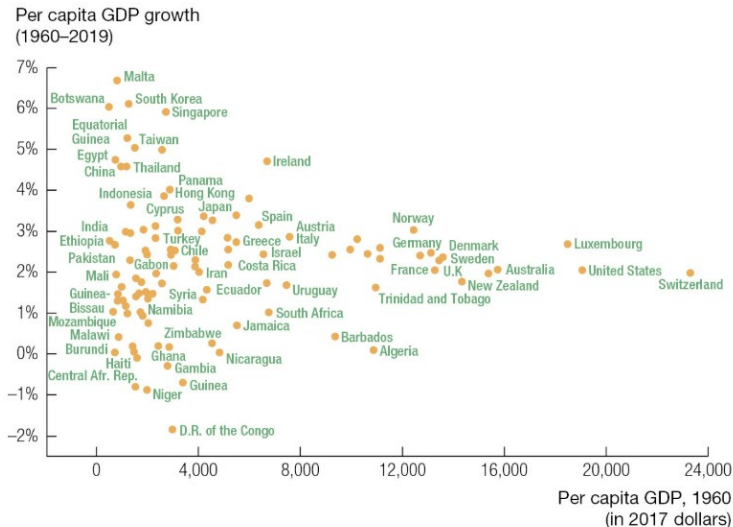
Why might the steady state change over time?

- Permanent change in savings rate within a country – if we save more, our steady state capital/output will increase over time, if we save less it will decrease
- Permanent shocks that affect the depreciation rate – better quality capital might decrease our depreciation rate, increasing our steady state capital/output, things like worse long-term climate that causes faster depreciation of capital could decrease our steady state capital/output
- Short-term shocks to capital (for example, natural disaster, war) can bump us up/down the model in the short-run, but if our savings rate and depreciation rate remain the same, we will return back to our steady-state in the long run

Growth across countries

- Empirically, different countries have different investment rates:
- For example, investment in South Korea has averaged 35% of GDP, investment in United States has been around 25%
- Empirically, we observe that investment rates in the richest countries average 25-30% while investment rates in poor countries are around 7%
 - Based on our model, differences in investment rates explain a factor of 2 times the differences in income, while 37.5 is explained by total factor productivity differences (!)
- If countries have similar growth rates despite very different output/GDP per capita, the Solow model suggests that all countries with similar growth rates are in a similar position relative to their steady state

Growth Rates Across Countries



Lessons from the Solow Model

- In the long run, diminishing returns to capital accumulation cause investments to slow down
- Eventually, depreciation and new investment offset each other and we reach steady state
- Capital accumulation cannot serve as the engine of long-run economic growth

PSET Q3 – The Solow Model with Population Growth

$$y_t = \bar{A}k_t^{1/3}, \quad (5.14)$$

$$\Delta k_{t+1} = \bar{s}y_t - (\bar{n} + \bar{d})k_t. \quad (5.15)$$

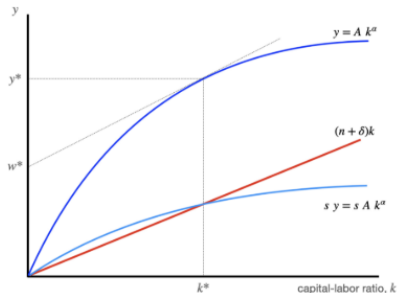
PSET Questions

3. [16 points total] So how low can we go, Solow? A capital response to immigration?

In the standard neoclassical growth model of Robert Solow without technological growth, consumers supply labor inelastically and save out of income at rate s . There are constant returns to scale in labor and capital, and productivity A is fixed and not changing. Capital depreciates at rate δ , and labor increases at the rate of population growth, n . When output is Cobb-Douglas,

capital per worker is	$k = K/L$	also called the "capital-labor ratio"
output per worker is	$y = A k^\alpha$	and typically $\alpha = 1/3$
saving per worker is	$s y = s A k^\alpha$	
depreciation per worker is	$(n + \delta)k$	

The Solow diagram below shows how capital per worker reaches a *steady state* at k^* , where depreciation per worker equals saving per worker: $(n + \delta)k = s y$, so capital per worker is unchanging there. The steady state is shown by the vertical dashed line in the diagram below.



PSET Questions

- (a) [2 points] If the capital per worker is in steady state at k^* and not changing over time, what is true about the wage w^* , which is shown in the graph on the y-axis?

PSET Questions

Suppose that the economy is in steady state, and then the government carries out a large removal of immigrant workers, one that increases capital per worker k by about 25%. The policy removes workers but leaves capital K and productivity, A , unchanged.

- (b) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n , δ , s , A , and α change?]

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- (c) [2 points] A higher level of capital per worker will produce a higher wage, because workers are more productive. And removing any type of worker will raise capital per worker. But is that change temporary or permanent? Why?

PSET Questions

Now suppose that while in steady state, the government instead decides to change **immigration policy**, reducing number of refugee and work visas each year, which reduces the rate of population growth, n , to a new level $n' < n$. (Assume there is no removal policy.)

- (d) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n , δ , s , A , and α change?]
- (e) [4 points] What would have happened in the graph in part (d) if immigration enhanced productivity, A , so that reducing immigration reduced the level of productivity? Explain.